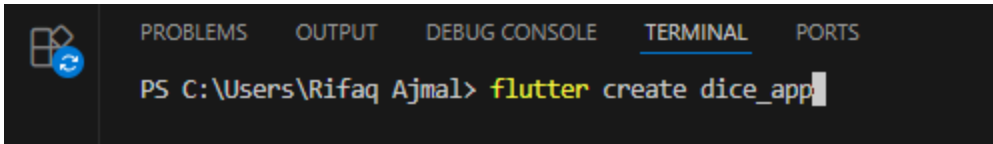


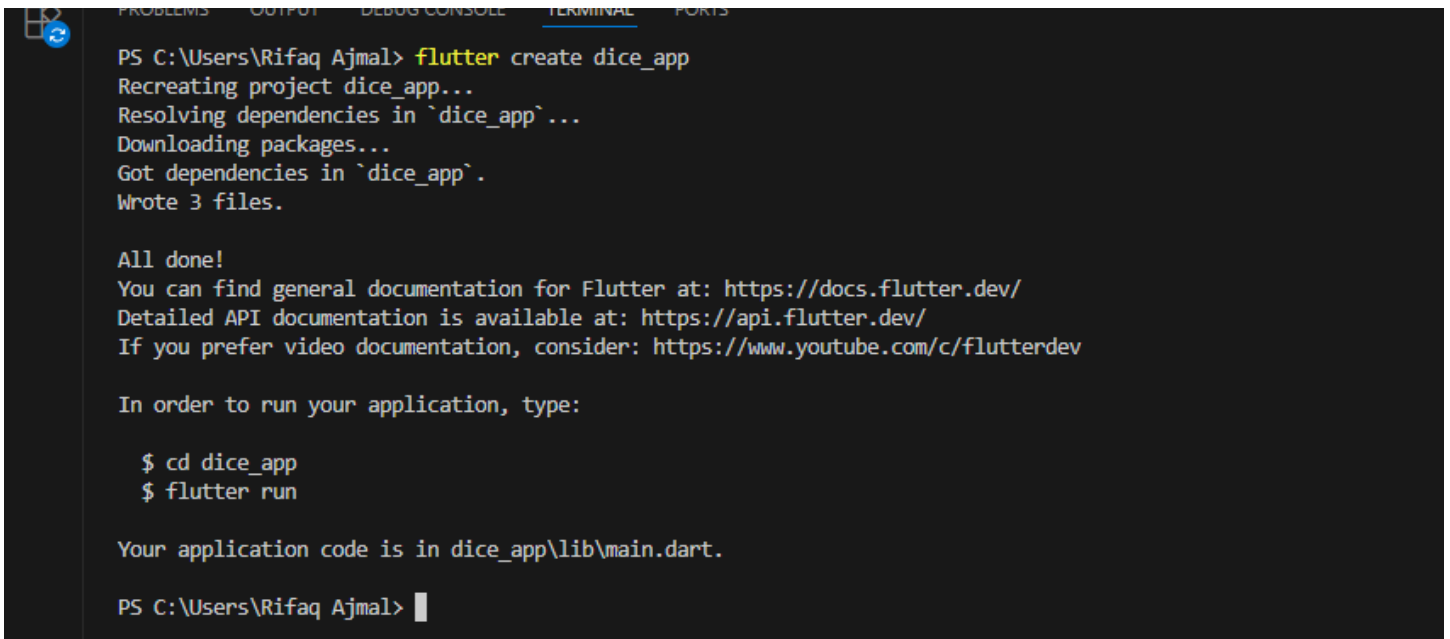
Dice App + Light ON/OFF App

Task1 : Dice App

Step 1 — Create Project



```
PS C:\Users\Rifaq Ajmal> flutter create dice_app
```



```
PS C:\Users\Rifaq Ajmal> flutter create dice_app
Recreating project dice_app...
Resolving dependencies in `dice_app`...
Downloading packages...
Got dependencies in `dice_app`.
Wrote 3 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

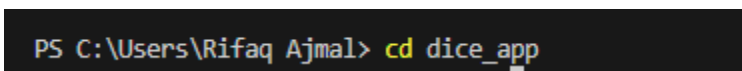
In order to run your application, type:

$ cd dice_app
$ flutter run

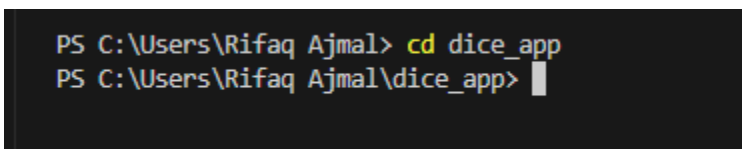
Your application code is in dice_app\lib\main.dart.

PS C:\Users\Rifaq Ajmal>
```

Go inside:



```
PS C:\Users\Rifaq Ajmal> cd dice_app
```

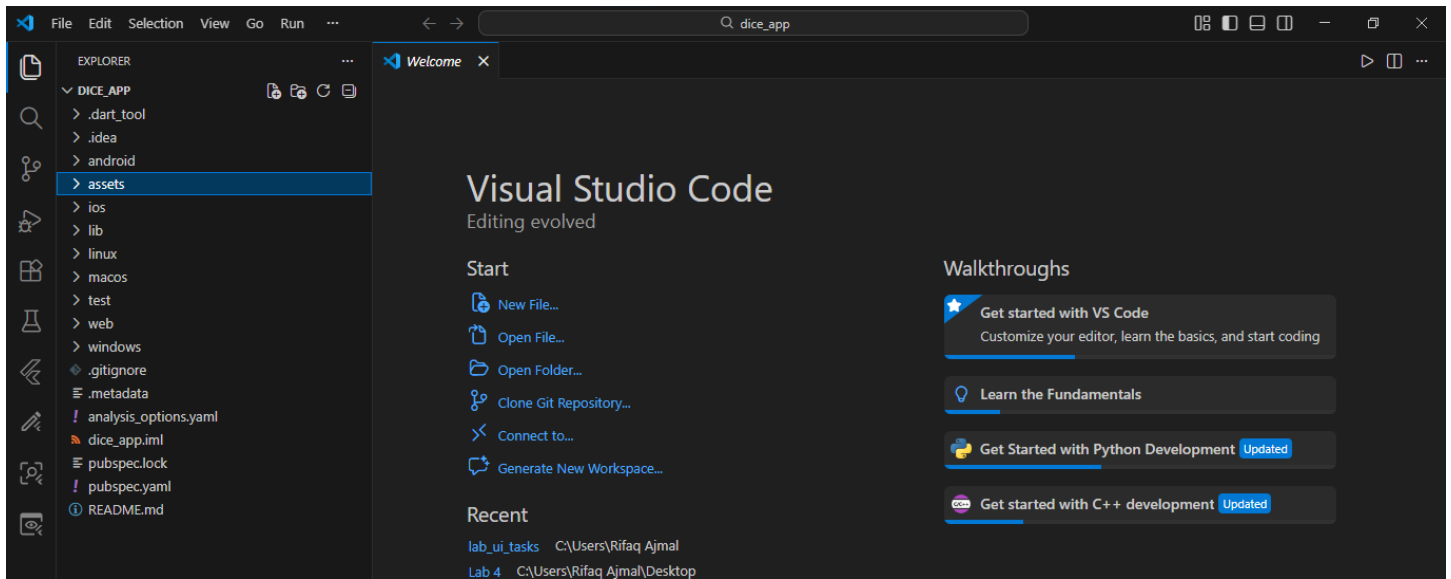


```
PS C:\Users\Rifaq Ajmal> cd dice_app
PS C:\Users\Rifaq Ajmal\dice_app>
```

Step 2: Open this folder in Visual Studio Code

file → Open Folder → dice_app

Step 3— Create Assets Folder



STEP 4 — Get Dice Images/ Download

STEP 5 — Move Images

Drag all images into:

`dice_app/assets/`

Download dice images, rename, add to the folder – assets



STEP 6 — Edit pubspec.yaml

Open file:

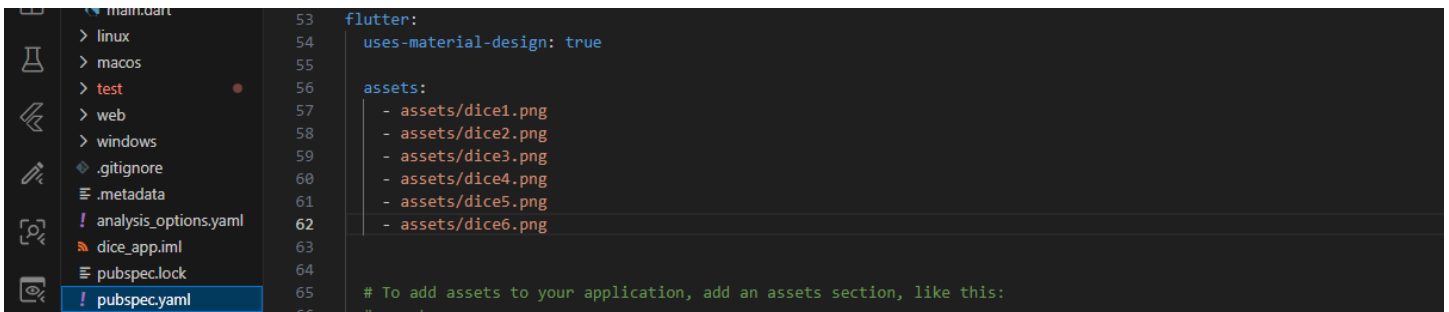
pubspec.yaml

Find:

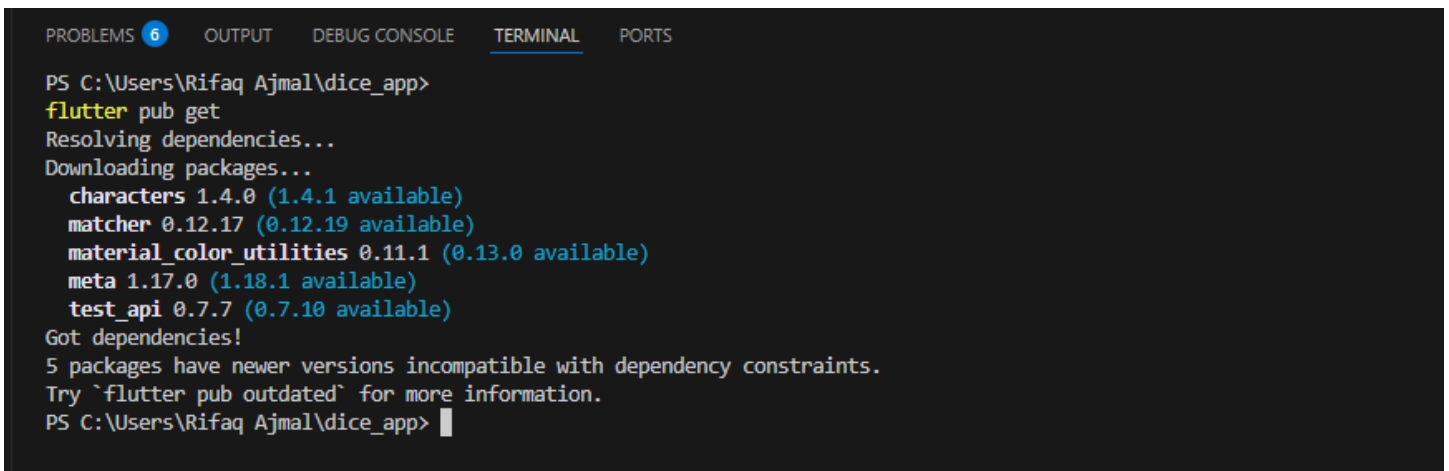
flutter:

We'Make it like this:

```
flutter:
  assets:
    - assets/dice1.png
    - assets/dice2.png
    - assets/dice3.png
    - assets/dice4.png
    - assets/dice5.png
    - assets/dice6.png
```



STEP 8 — Running

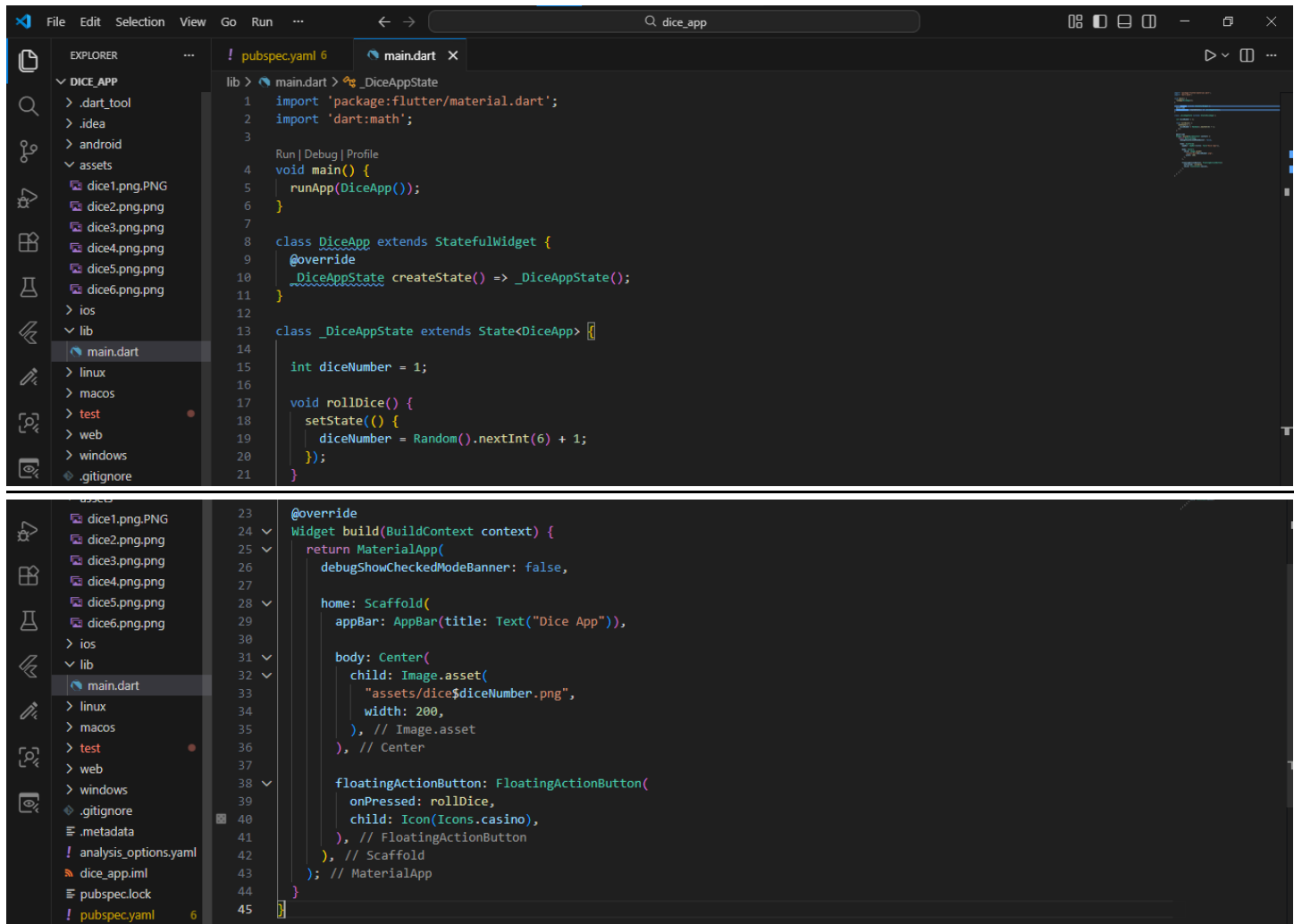


STEP 8 — Add Code

lib/main.dart

Delete everything and paste:

Code



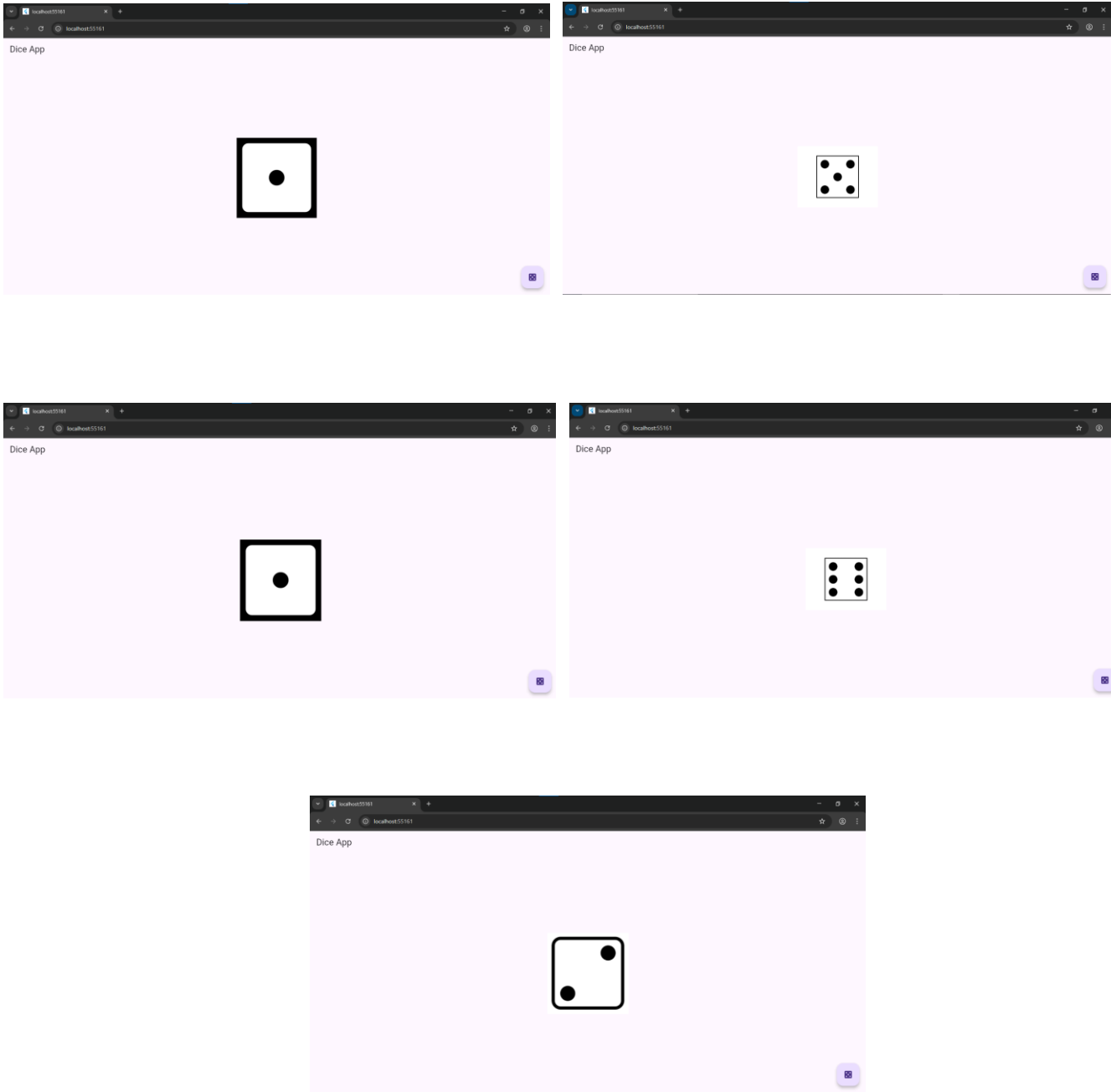
The screenshot shows an IDE with two panels. The top panel displays the file explorer on the left, showing a project named 'DICE_APP' with various assets and a 'lib' directory containing 'main.dart'. The main editor shows the code for 'main.dart', which includes imports for 'package:flutter/material.dart' and 'dart:math', a 'main' function that runs 'DiceApp()', and a 'DiceApp' class that extends 'StatefulWidget'. The 'DiceApp' class has an '@override' method 'createState()' that returns a '_DiceAppState' object. The '_DiceAppState' class extends 'State<DiceApp>' and contains a 'diceNumber' variable initialized to 1, a 'rollDice()' method that updates the 'diceNumber' to a random value between 1 and 6, and a 'build' method that returns a 'MaterialApp' widget. The 'MaterialApp' widget has a 'Scaffold' with an 'AppBar' titled 'Dice App', a 'body' containing an 'Image.asset' of a dice, and a 'floatingActionButton' with an 'onPressed' callback 'rollDice()' and an 'Icon(Icons.casino)'.

```
lib > main.dart > _DiceAppState
1  import 'package:flutter/material.dart';
2  import 'dart:math';
3
4  Run | Debug | Profile
5  void main() {
6    runApp(DiceApp());
7  }
8
9  class DiceApp extends StatefulWidget {
10   @override
11   _DiceAppState createState() => _DiceAppState();
12 }
13
14 class _DiceAppState extends State<DiceApp> {
15
16   int diceNumber = 1;
17
18   void rollDice() {
19     setState(() {
20       diceNumber = Random().nextInt(6) + 1;
21     });
22   }
23
24   @override
25   Widget build(BuildContext context) {
26     return MaterialApp(
27       debugShowCheckedModeBanner: false,
28
29       home: Scaffold(
30         appBar: AppBar(title: Text("Dice App")),
31
32         body: Center(
33           child: Image.asset(
34             "assets/dice$diceNumber.png",
35             width: 200,
36           ), // Image.asset
37         ), // Center
38
39         floatingActionButton: FloatingActionButton(
40           onPressed: rollDice,
41           child: Icon(Icons.casino),
42         ), // FloatingActionButton
43       ), // Scaffold
44     ); // MaterialApp
45   }
46 }
```

STEP 9 — Run App

flutter run -d chrome

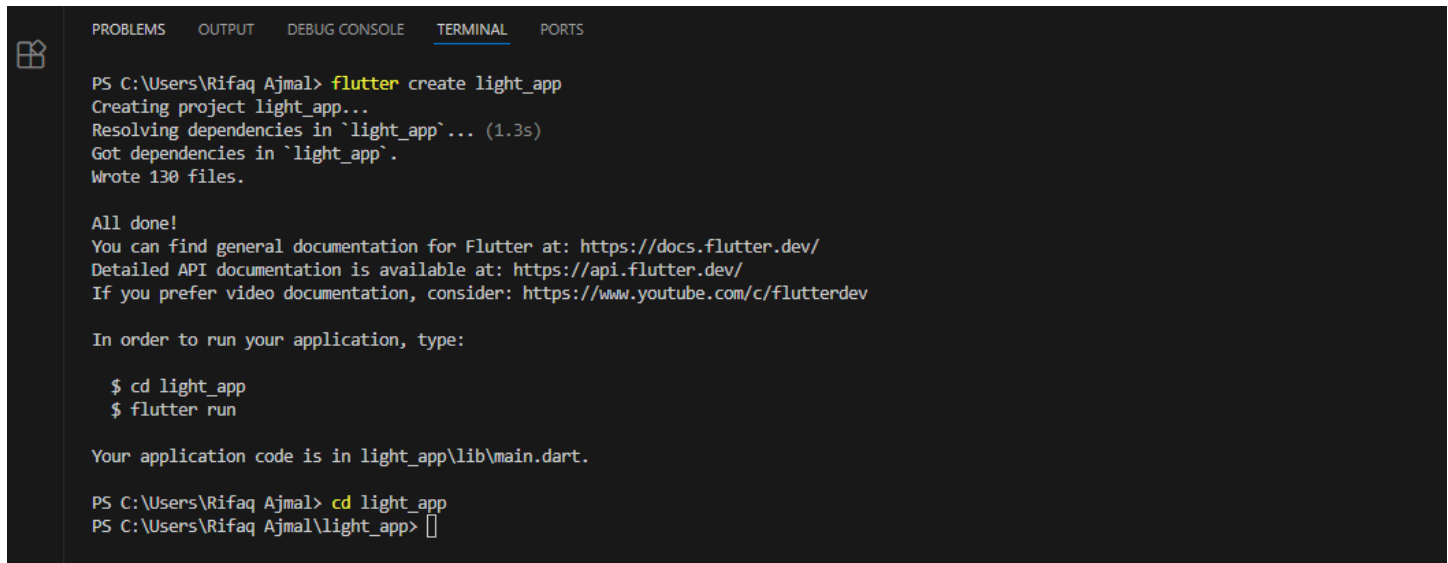
Result



Task 2 — Light ON/OFF

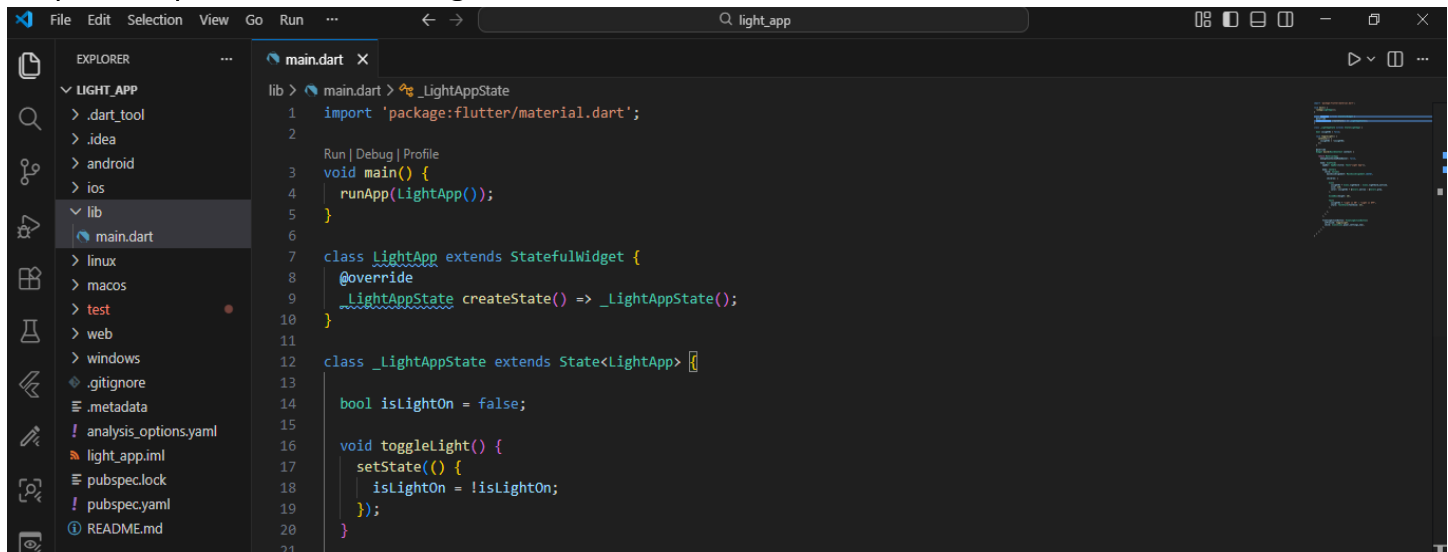
Step 1 — Create New Project

```
flutter create light_app  
cd light_app
```

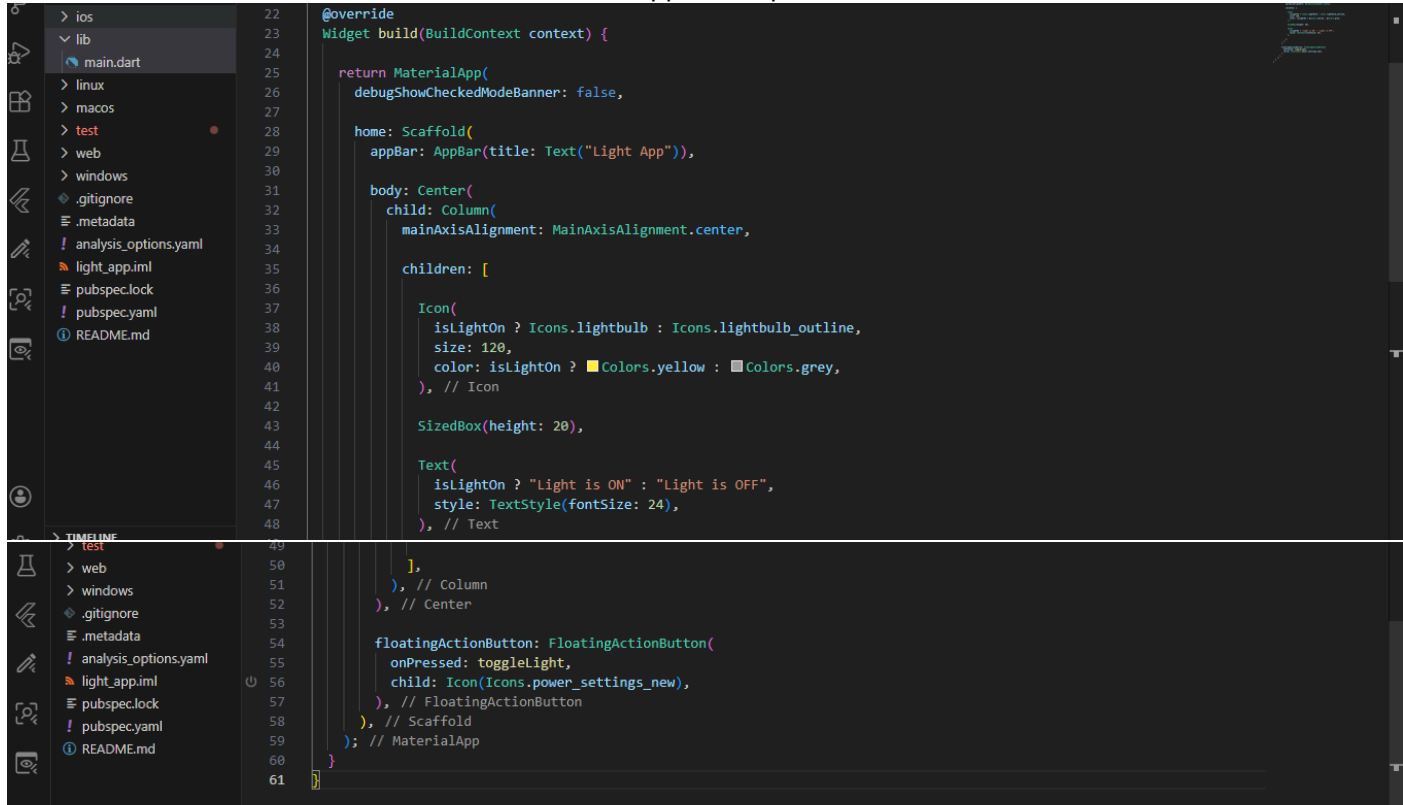


```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\Rifaq Ajmal> flutter create light_app  
Creating project light_app...  
Resolving dependencies in `light_app`... (1.3s)  
Got dependencies in `light_app`.  
Wrote 130 files.  
  
All done!  
You can find general documentation for Flutter at: https://docs.flutter.dev/  
Detailed API documentation is available at: https://api.flutter.dev/  
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev  
  
In order to run your application, type:  
  
$ cd light_app  
$ flutter run  
  
Your application code is in light_app\lib\main.dart.  
  
PS C:\Users\Rifaq Ajmal> cd light_app  
PS C:\Users\Rifaq Ajmal\light_app>
```

Step 2 — Open lib folder change code



```
File Edit Selection View Go Run ...  
light_app  
EXPLORER  
LIGHT_APP  
  .dart_tool  
  .idea  
  android  
  ios  
  lib  
    main.dart  
  linux  
  macos  
  test  
  web  
  windows  
  .gitignore  
  .metadata  
  analysis_options.yaml  
  light_app.iml  
  pubspec.lock  
  pubspec.yaml  
  README.md  
main.dart X  
lib > main.dart > _LightAppState  
1 import 'package:flutter/material.dart';  
2  
3 Run | Debug | Profile  
4 void main() {  
5   runApp(LightApp());  
6 }  
7  
8 class LightApp extends StatefulWidget {  
9   @override  
10  _LightAppState createState() => _LightAppState();  
11 }  
12  
13 class _LightAppState extends State<LightApp> {  
14  
15   bool isLightOn = false;  
16  
17   void toggleLight() {  
18     setState(() {  
19       isLightOn = !isLightOn;  
20     });  
21   }  
22 }
```



```

@override
Widget build(BuildContext context) {
  return MaterialApp(
    debugShowCheckedModeBanner: false,
    home: Scaffold(
      appBar: AppBar(title: Text("Light App")),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Icon(
              isLightOn ? Icons.lightbulb : Icons.lightbulb_outline,
              size: 120,
              color: isLightOn ? Colors.yellow : Colors.grey,
            ), // Icon

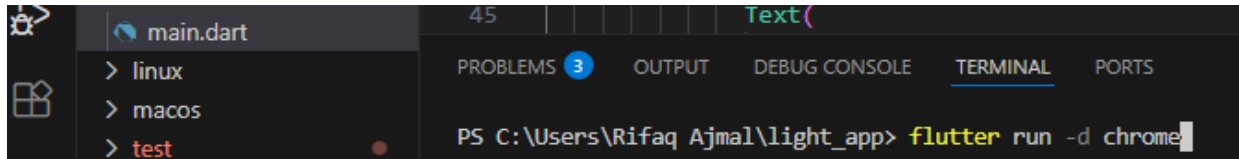
            SizedBox(height: 20),

            Text(
              isLightOn ? "Light is ON" : "Light is OFF",
              style: TextStyle(fontSize: 24),
            ), // Text
          ],
        ), // Column
      ), // Center
      floatingActionButton: FloatingActionButton(
        onPressed: toggleLight,
        child: Icon(Icons.power_settings_new),
      ), // FloatingActionButton
    ), // Scaffold
  ); // MaterialApp
}

```

Step 3 — Run

flutter run -d chrome



```

PS C:\Users\Rifaq Ajmal\light_app> flutter run -d chrome

```

Result

